

Homework 7

Ravi Kini

December 11, 2025

Exercise 1

Let $n = 4861$ and $x = 11$. Since $4860 = 2^{12} + 2^9 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2$, we see that:

k	$11^{2^k} \bmod 4861$	k	$11^{2^k} \bmod 4861$
0	11	7	$4240^2 \bmod 4861 = 1622$
1	$11^2 \bmod 4861 = 121$	8	$1622^2 \bmod 4861 = 1083$
2	$121^2 \bmod 4861 = 58$	9	$1083^2 \bmod 4861 = 1388$
3	$58^2 \bmod 4861 = 3364$	10	$1388^2 \bmod 4861 = 1588$
4	$3364^2 \bmod 4861 = 88$	11	$1588^2 \bmod 4861 = 3746$
5	$88^2 \bmod 4861 = 2883$	12	$3746^2 \bmod 4861 = 3670$
6	$2883^2 \bmod 4861 = 4240$		

$$\begin{aligned} 11^{4860} &= 11^{2^{12}+2^9+2^7+2^6+2^5+2^4+2^3+2^2} \bmod 4861 \\ &= 11^{2^{12}} \cdot 11^{2^9} \cdot 11^{2^7} \cdot 11^{2^6} \cdot 11^{2^5} \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \bmod 4861 \\ &= 3670 \cdot 1388 \cdot 11^{2^7} \cdot 11^{2^6} \cdot 11^{2^5} \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \bmod 4861 \\ &= 4493 \cdot 1622 \cdot 11^{2^6} \cdot 11^{2^5} \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \bmod 4861 \\ &= 1007 \cdot 4240 \cdot 11^{2^5} \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \bmod 4861 \\ &= 1722 \cdot 2883 \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \bmod 4861 \\ &= 1445 \cdot 88 \cdot 11^{2^3} \cdot 11^{2^2} \bmod 4861 \\ &= 774 \cdot 3364 \cdot 11^{2^2} \bmod 4861 \\ &= 3101 \cdot 58 \bmod 4861 \\ &= 1 \end{aligned}$$

Then, since $4860/2 = 2^{11} + 2^8 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1$:

$$\begin{aligned}
11^{4860/2} &= 11^{2^{11}+2^8+2^6+2^5+2^4+2^3+2^2+2^1} \pmod{4861} \\
&= 11^{2^{11}} \cdot 11^{2^8} \cdot 11^{2^6} \cdot 11^{2^5} \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \cdot 11^{2^1} \pmod{4861} \\
&= 3746 \cdot 1083 \cdot 11^{2^6} \cdot 11^{2^5} \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \cdot 11^{2^1} \pmod{4861} \\
&= 2844 \cdot 4240 \cdot 11^{2^5} \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \cdot 11^{2^1} \pmod{4861} \\
&= 3280 \cdot 2883 \cdot 11^{2^4} \cdot 11^{2^3} \cdot 11^{2^2} \cdot 11^{2^1} \pmod{4861} \\
&= 1595 \cdot 88 \cdot 11^{2^3} \cdot 11^{2^2} \cdot 11^{2^1} \pmod{4861} \\
&= 4252 \cdot 3364 \cdot 11^{2^2} \cdot 11^{2^1} \pmod{4861} \\
&= 2666 \cdot 58 \cdot 11^{2^1} \pmod{4861} \\
&= 3937 \cdot 121 \pmod{4861} \\
&= 4860
\end{aligned}$$

Since $4860/3 = 2^{10} + 2^9 + 2^6 + 2^4 + 2^2$:

$$\begin{aligned}
11^{4860/3} &= 11^{2^{10}+2^9+2^6+2^4+2^2} \pmod{4861} \\
&= 11^{2^{10}} \cdot 11^{2^9} \cdot 11^{2^6} \cdot 11^{2^4} \cdot 11^{2^2} \pmod{4861} \\
&= 1588 \cdot 1388 \cdot 11^{2^6} \cdot 11^{2^4} \cdot 11^{2^2} \pmod{4861} \\
&= 2111 \cdot 4240 \cdot 11^{2^4} \cdot 11^{2^2} \pmod{4861} \\
&= 1539 \cdot 88 \cdot 11^{2^2} \pmod{4861} \\
&= 4185 \cdot 58 \pmod{4861} \\
&= 4541
\end{aligned}$$

Since $4860/5 = 2^9 + 2^8 + 2^7 + 2^6 + 2^3 + 2^2$:

$$\begin{aligned}
11^{4860/5} &= 11^{2^9+2^8+2^7+2^6+2^3+2^2} \pmod{4861} \\
&= 11^{2^9} \cdot 11^{2^8} \cdot 11^{2^7} \cdot 11^{2^6} \cdot 11^{2^3} \cdot 11^{2^2} \pmod{4861} \\
&= 1388 \cdot 1083 \cdot 11^{2^7} \cdot 11^{2^6} \cdot 11^{2^3} \cdot 11^{2^2} \pmod{4861} \\
&= 1155 \cdot 1622 \cdot 11^{2^6} \cdot 11^{2^3} \cdot 11^{2^2} \pmod{4861} \\
&= 1925 \cdot 4240 \cdot 11^{2^3} \cdot 11^{2^2} \pmod{4861} \\
&= 381 \cdot 3364 \cdot 11^{2^2} \pmod{4861} \\
&= 3241 \cdot 58 \pmod{4861} \\
&= 3260
\end{aligned}$$

Since $11^{4860} = 1 \pmod{4861}$, and for all prime factors $q \in \{2, 3, 5\}$ of 4860, $11^{4860/q} \neq 1 \pmod{4861}$, 4861 is prime.

Exercise 2

Since E (4) and T (19) are the most common letters in the English alphabet, they are the most likely to be mapped to W (22) and B (1). Then, by the affine transformation:

$$4a + b = 22 \bmod 26 \quad 19a + b = 1 \bmod 26$$

It follows that:

$$15a = 5 \bmod 26$$

$$a = 15^{-1} \cdot 5 = 7 \cdot 5 = 11 \bmod 26 \quad 4 \cdot 11 + b = 22 \bmod 26$$

$$b = 22 - 4 \cdot 11 = 4 \bmod 26$$

We see that $a = 11, b = 4$.

Exercise 3

Using modular exponentiation with $p = 29$, $e = 17$, $d = 5$ since $\phi(29) = 28$ and $17 \cdot 5 = 1 \bmod 28$. If $C = (13\ 11\ 11\ 02)$

$$\begin{aligned} P &= C^d \bmod p \\ &= (13\ 11\ 11\ 02)^5 \bmod 29 \\ &= (06\ 14\ 14\ 03) \end{aligned}$$

The plaintext $(06\ 14\ 14\ 03)$ translates to the message GOOD.

Exercise 4

BEST translates to (01 04 18 19). Using RSA encryption with $n = 77$ and $e = 7$:

$$\begin{aligned} C &= P^e \bmod n \\ &= (01\ 04\ 18\ 19)^7 \bmod 77 \\ &= (01\ 60\ 39\ 68) \end{aligned}$$

The ciphertext is (01 60 39 68).

Exercise 5

Using RSA encryption with $n = 77$, $e = 43$, $d = 7$ since $\phi(77) = 60$ and $43 \cdot 7 = 1 \pmod{60}$. If $C = (01\ 49\ 49\ 10)$

$$\begin{aligned} P &= C^d \pmod{n} \\ &= (01\ 49\ 49\ 10)^7 \pmod{77} \\ &= (01\ 14\ 14\ 10) \end{aligned}$$

The plaintext $(01\ 14\ 14\ 10)$ translates to the message BOOK.